Física Aplicada **Eng. Informática**

2024/2025

Formulário

$|\vec{F}| = k\,\frac{q_1\,q_2}{r^2}$; $\vec{F} = q\,\vec{E}$; $|\vec{E}| = k\,\frac{q}{r^2}$; $|\vec{E}| = \frac{\Delta V}{d}$; $|\vec{E}| = \frac{\sigma}{2\varepsilon_0}$; $(\sigma = \frac{q}{A})$

$V = k\frac{q}{r}$; $E_p = q\,V$; $W_{A\to B} = q\,(V_A - V_B)$; $\Delta E_C = -\Delta E_P$; $E_C = \frac{1}{2}\,m\,v^2$; $W = \vec{F}\cdot\vec{d}$

$i = \frac{dq}{dt}$; $I = \frac{\Delta Q}{\Delta t}$; $V = R\,I$; $R = \rho\frac{l}{A}$; $\rho - \rho_0 = -\alpha(T - T_0)$; $R_{eq} = \sum_i R_i$; $\frac{1}{R_{eq}} = \sum_i \frac{1}{R_i}$

$\mathcal{E} = \frac{E_{transformada}}{Q}$; $P = \frac{\Delta E}{\Delta t}$; $P = VI$; $\sum_i V_i = 0$; $\sum_i I_i = 0$

$V = \frac{Q}{C}$; $C = \varepsilon\,\frac{A}{d}$; $\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$; $E = \frac{1}{2}Q\,V$; $C_{eq} = \sum_i C_i$; $1/C_{eq} = \sum_i 1/C_i$

$q(t) = q_{total}\left(1 - \mathrm{e}^{\left(-\frac{t}{\tau}\right)}\right)$; $q(t) = q_{total}\left(e^{-\frac{t}{\tau}}\right)$; $\tau = R_{eq}C$; $i(t) = I_0 e^{-t/\tau}$

$V(t) = V_{máxima}\left(1 - \mathrm{e}^{\left(-\frac{t}{\tau}\right)}\right)$; $V(t) = V_{máxima}\left(e^{-\frac{t}{\tau}}\right)$;

$\vec{F}_{em} = q\vec{E} + q\vec{v}\times\vec{B}$ $\vec{F}_m = q\,\vec{v}\times\vec{B}$, $d\vec{F} = I\,d\vec{l}\times\vec{B}$; $F_c = m\frac{v^2}{R}$

$f = \frac{1}{T}$; $v = \frac{dx}{dt}$; $\omega = \frac{v}{R}$; $\omega = 2\pi f$; $\sum\vec{F}_i = m\vec{a}$

$\vec{B} = \frac{\mu_0}{4\pi}\,\frac{q\,\vec{v}\times\hat{r}}{r^2}$; $\vec{B} = \frac{\mu_0}{4\pi}\,\frac{I\,d\vec{l}\times\hat{r}}{r^2}$; $B = \mu_0\,n\,I$ (solenoide); $B = \frac{\mu_0}{4\pi}\frac{2I}{R}$ (fio recto e longo);

$B = \frac{\mu_0}{2}\frac{I}{R}$ *(no centro duma espira)*; $\Phi_m = \int\vec{B}\cdot d\vec{A}$ $\Phi_m = N\,B\,A\,cos\theta$;

$\varepsilon_{induzida} = -\frac{d\Phi}{dt}$; $\varepsilon_{induzida} = v\,B\,l$

$\frac{V_P}{V_S} = \frac{N_P}{N_S}$; $\frac{i_P}{i_S} = \frac{N_S}{N_P}$; $P_P = P_S$

$\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}$; $I = \frac{P}{área} = \frac{Energia}{tempo\times área}$; $I = |\vec{S}|_{médio} = \frac{1}{2\mu_0}E_m B_m$

$E = E_m\,sen(\omega\,t - k\,x)$; $B = B_m\,sen(\omega\,t - k\,x)$; $c = \frac{E_m}{B_m}$

$c = \lambda\,f$; $\omega = 2\pi\,f$; $k = \frac{2\pi}{\lambda}$

$I_{transmitido} = I_{incidente} cos^2\theta;$ $\quad n = \frac{c}{v};$ $\quad n_{ar} = 1$

$\theta_{incidente} = \theta_{reflectido}\,;$ $\quad n_1 . \sin\theta_{incidente} = n_2 . \sin\theta_{refractado}\,;$

$\theta_{critico} = \sin^{-1}(^{n_{transmitido}}/_{n_{incidente}})\,;$ $\quad A.N. = \sin\phi = \sqrt{n^2_{núcleo} - n^2_{bainha}}$

$Q = c\, m\, \Delta T;$ $\quad Q = m\, \lambda;$ $\quad \Delta l = l\, \alpha\, \Delta T;$ $\quad \Delta V = V\, \beta\, \Delta T;$ $\quad \lambda_{max}\, T = 2{,}898 \times 10^{-3}$

$q = \frac{\Delta Q}{\Delta t};$ $\quad q = k\, A\, \frac{\Delta T}{\Delta x};$ $\quad q = h\, A\, \Delta T;$ $\quad P = \varepsilon\, \sigma\, A\, T^4;$ $\quad q = \varepsilon\, \sigma\, A\, (T_f^4 - T_{viz}^4)$

$q = kA\frac{\partial T}{\partial x};$ $\quad q = \frac{\partial}{\partial \mathrm{x}}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial \mathrm{y}}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial \mathrm{z}}\left(k\frac{\partial T}{\partial z}\right);$ $\quad R = \frac{\Delta x}{k\, A};$ $\quad R = \frac{1}{h\, A};$ $\quad \Delta E_{int} = Q + W$

$\vec{r} = \vec{r}_0 + \vec{v}_0 \cdot t + \frac{1}{2}\vec{a} \cdot t^2$

<u>Constantes Físicas</u>:

$\sigma = 5{,}6697 \times 10^{-8}$ W m^{-2} K^{-4} (Stefan-Boltzmann)

k = 1,381 x 10^{-3} J K^{-1} (Boltzmann)

N_A = 6,022 x 10^{23} mol^{-1} (número de Avogadro)

R = 8,32 J mol^{-1} K^{-1} (constante universal dos gases perfeitos)

$q_{elementar}$ = 1,60217653 × 10^{-19} C

m_e = 9,109 x 10^{-31} kg (massa do electrão)

m_p = 1.673 x 10^{-27} kg (massa do protão)

k = 8,99 x 10^9 N m^2 C^{-2} (constante de Coulomb)

$\varepsilon_0 = 8{,}85 \times 10^{-12}$ C^2 N^{-1} m^{-2} (permissividade eléctrica no vácuo)

μ_0 = 4 π x 10 $^{-7}$ N/A^2 (permeabilidade magnética no vácuo)

c = 3,0 x 10^8 m/s

G = 6,67 x 10^{-11} N m^2 kg^{-2}

g = 9,81 m/s^2